STACKS, DIFFERENTIAL GRADED SHEAVES: LEMMA-PUSHFORWARD-AGREES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5377-5396 of sources/stacks-sdga/pinned-original-source.tex; proof span is lines 5398-5448, the complete author proof (49 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 11 of queue_sdga.json (pb=2091, stmt 5377-5396, proof 5398-5448).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0FTP: sources/stacks-sdga/pinned-original-source.tex, source lines 5135–5165. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FTU: sources/stacks-sdga/pinned-original-source.tex, source lines 5324–5338. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0FRT: sources/stacks-sdga/pinned-original-source.tex, source lines 2137–2225. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07A5: sources/stacks-sdga/pinned-original-source.tex, source lines 5448–5448. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $(f, f^\sharp) : (Sh(\mathcal{C}), \mathcal{O}_\mathcal{C}) \to (Sh(\mathcal{D}), \mathcal{O}_\mathcal{D})$ be a morphism of ringed topoi. Let $\mathcal{A}$ be a differential graded $\mathcal{O}_\mathcal{C}$-algebra. Let $\mathcal{B}$ be a differential graded $\mathcal{O}_\mathcal{D}$-algebra. Let $\varphi : \mathcal{B} \to f_*\mathcal{A}$ be a homomorphism of differential graded $\mathcal{O}_\mathcal{D}$-algebras.*

Date: stacks-project:sdga.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

The diagram

$$\begin{array}{ccc} D(\mathcal{A}, d) & \xrightarrow{forget} & D(\mathcal{O}_\mathcal{C}) \\ {\scriptstyle Rf_*}\downarrow & & \downarrow{\scriptstyle Rf_*} \\ D(\mathcal{B}, d) & \xrightarrow{forget} & D(\mathcal{O}_\mathcal{D}) \end{array}$$

commutes.

Proof. Besides identifying some categories, this lemma follows immediately from Lemma 0FTU.

We may view $(\mathcal{O}_\mathcal{C}, 0)$ as a differential graded $\mathcal{O}_\mathcal{C}$-algebra by placing $\mathcal{O}_\mathcal{C}$ in degree 0 and endowing it with the zero differential. It is clear that we have

$$Mod(\mathcal{O}_\mathcal{C}, 0) = \mathrm{Comp}(\mathcal{O}_\mathcal{C}) \quad \text{and} \quad D(\mathcal{O}_\mathcal{C}, 0) = D(\mathcal{O}_\mathcal{C})$$

Via this identification the forgetful functor $Mod(\mathcal{A}, \mathrm{d}) \to \mathrm{Comp}(\mathcal{O}_\mathcal{C})$ is the "push-forward" $\mathrm{id}_{\mathcal{C},*}$ defined in Section 0FRT corresponding to the identity morphism $\mathrm{id}_\mathcal{C} : (\mathcal{C}, \mathcal{O}_\mathcal{C}) \to (\mathcal{C}, \mathcal{O}_\mathcal{C})$ of ringed topoi and the map $(\mathcal{O}_\mathcal{C}, 0) \to (\mathcal{A}, \mathrm{d})$ of differential graded $\mathcal{O}_\mathcal{C}$-algebras. Since $\mathrm{id}_{\mathcal{C},*}$ is exact, we immediately see that

$$R\mathrm{id}_{\mathcal{C},*} = forget : D(\mathcal{A}, \mathrm{d}) \longrightarrow D(\mathcal{O}_\mathcal{C}, 0) = D(\mathcal{O}_\mathcal{C})$$

The exact same reasoning shows that

$$R\mathrm{id}_{\mathcal{D},*} = forget : D(\mathcal{B}, \mathrm{d}) \longrightarrow D(\mathcal{O}_\mathcal{D}, 0) = D(\mathcal{O}_\mathcal{D})$$

Moreover, the construction of $Rf_* : D(\mathcal{O}_\mathcal{C}) \to D(\mathcal{O}_\mathcal{D})$ of Cohomology on Sites, Section 07A5 agrees with the construction of $Rf_* : D(\mathcal{O}_\mathcal{C}, 0) \to D(\mathcal{O}_\mathcal{D}, 0)$ in Definition 0FTP as both functors are defined as the right derived extension of pushforward on underlying complexes of modules. By Lemma 0FTU we see that both $Rf_* \circ R\mathrm{id}_{\mathcal{C},*}$ and $R\mathrm{id}_{\mathcal{D},*} \circ Rf_*$ are the derived functors of $f_* \circ forget = forget \circ f_*$ and hence equal by uniqueness of adjoints. □